



*Synergies between HEIs and the Labour Market for Developing Sustainability
Capabilities using Theory of Change and AI*

WP4 – Curriculum Content Development

PR7 – Pilot Testing Guidelines & Evaluation Questionnaire

Developed by Instituto Politécnico de Santarém (SPU)

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Table of Contents

1. Introduction
 2. Pilot Testing – Theoretical Framework
 3. Recruitment of Participants
 4. Implementation
 - 4.1 Pilot Testing Deadlines
 - 4.2 Course Content to be Evaluated
 - 4.3 Evaluation Methodology
 - 4.4 Reporting of Findings
 - 4.5 Course Improvement Process
 - 4.6 Instructions for Completing the Evaluation Survey
 5. References
- Annex I – Pilot Testing National Report Template
- Annex II – WP4 Evaluation Questionnaire

1. Introduction

This document establishes the methodological guidelines for the pilot testing of the GreenToC semester course on green transformation strategy development for the labour market (PR7), conducted within the framework of Work Package 4 (WP4 – Curriculum Content Development) of the GreenToC Erasmus+ project (KA220-HED).

The GreenToC project (Synergies between Higher Education Institutions and the Labour Market for Developing Sustainability Capabilities using Theory of Change and AI) is an Erasmus+ Cooperation Partnership in Higher Education. Its overarching aim is to upskill HEI students in the development of green transformation strategies through leveraging digital and AI tools, and through their practical engagement with organisations in the labour market.

The present pilot testing guidelines have been adapted from the methodological framework developed for the hVET project (Breathe the Future – Hybrid Education in VET, 2021-2-PL01-KA220-VET-000050664) and recalibrated to address the specific objectives, participant profiles, course structure, and evaluation requirements of WP4 of the GreenToC project.

2. Pilot Testing – Theoretical Framework

A pilot test is a methodologically foundational procedure in educational research and programme evaluation. According to Crossman (2019), conducting a pilot study prior to full-scale implementation allows researchers and course designers to execute their programmes with greater methodological rigour, while simultaneously reducing the risk of errors and optimising resource allocation.

In the context of higher education curriculum development, pilot testing serves as a critical quality assurance mechanism (Fraser et al., 2018). Teijlingen and Hundley (2001) identify several core functions of pilot studies, including the collection of preliminary data, the identification of logistical and procedural challenges, and the calibration of instruments such as evaluation questionnaires.

The pilot testing phase of WP4 (PR7) will enable the GreenToC consortium to:

- Assess the scientific and pedagogical quality of the three course modules developed in PR6;
- Evaluate the usability and functionality of the GreenToC learning platform as a delivery mechanism;
- Gauge the effectiveness of the Theory of Change (ToC) methodology as an instructional framework for green strategy development;
- Assess the integration and utility of digital and AI tools within the learning process;
- Evaluate the practical experience component involving collaboration with labour market organisations;
- Collect evidence-based data to inform iterative course improvement prior to full-scale implementation.

Pilot studies may present limitations, including small sample sizes and potential sampling bias (Teijlingen & Hundley, 2001). The consortium has established minimum participant thresholds and standardised evaluation instruments to mitigate these risks.

3. Recruitment of Participants

The pilot testing of the GreenToC course involves three distinct participant categories, each fulfilling a specific role within the evaluation framework:

Target Participants per Partner Country

Participant Category	Profile & Role	Minimum per Country
HEI Students	Enrolled in the GreenToC semester course; engaged in both classroom and practical experience phases	10
Labour Market Representatives	Employees/managers from organisations participating in the practical experience component	5
Academics / Course Instructors	Teaching staff responsible for delivering and facilitating the course; trained on the GreenToC academic platform	1–3

The following evidence-based recruitment strategies are recommended for securing the participation of students and labour market organisations:

For Students:

- Formal course integration: embedding the GreenToC course as an elective or optional credited module within existing HEI curricula;
- Direct academic outreach: communications via faculty channels and departmental coordinators;
- Emphasis on competency development: highlighting transferable skills in sustainability, AI literacy, and strategic thinking;
- Professional development value: offering participation certificates and documentation of practical experience for student portfolios.

For Labour Market Organisations:

- Institutional partnerships: engaging organisations already connected to partner universities through internship or service-learning frameworks;
- Sector-specific outreach: targeting SMEs and public bodies with a declared interest in sustainability or ESG compliance;
- Value proposition communication: articulating the concrete benefit of receiving a tailored green transformation strategy at no cost to the organisation;
- Networking events: presenting GreenToC at sustainability-focused business forums or chambers of commerce.

4. Implementation

4.1 Pilot Testing Deadlines

The pilot testing of the GreenToC course is scheduled to take place between September 2026 and January 2027, in alignment with the WP4 timeline established in the project proposal. The table below presents the key activities and their corresponding deadlines:

Activity	Description	September 2026	October 2026
WP4_PT1	Recruitment of students & labour market representatives	1st Sep	
WP4_PT2	Pilot implementation – Phase 1: Classroom teaching (ToC & AI)	15th Sep	
WP4_PT3	Pilot implementation – Phase 2: Practical experience in labour market		1st Oct
WP4_PT4	Administration of evaluation questionnaire		15th Oct
WP4_PT5	Preparation of national pilot reports		31st Oct
WP4_PT6	Preparation of summative global report		

Note: The global summative report integrating all national reports will be finalised by March 2027 under the lead of IPSantarém (SPU).

4.2 Course Content to be Evaluated

During the pilot testing phase, participants will evaluate both the GreenToC learning platform and the content of the three course modules (PR6). The modules and their respective content foci are detailed below:

Module	Content Focus
Module 1: Green Transformation & Climate Change	Climate science; sustainability concepts; EU Green Deal; circular economy; environmental governance frameworks
Module 2: Theory of Change (ToC) Methodology	ToC principles; logical frameworks; outcome mapping; designing green transformation strategies for organisations
Module 3: Digital & AI Technologies for Green Strategies	AI-powered sustainability tools; digital platforms; data-driven decision making; developing green strategies with technology

The GreenToC platform can be accessed at: <https://project-spaces.eu/greentoc/>

4.3 Evaluation Methodology

The evaluation of the GreenToC course during the pilot phase will be conducted through a standardised multi-stakeholder questionnaire (Annex II), available in all partner languages. The questionnaire instrument is designed to capture quantitative data through Likert-scale items as well as qualitative data through open-ended questions, enabling a comprehensive mixed-methods analysis of course effectiveness.

The evaluation covers four dimensions:

- Platform usability and technical performance;
- Scientific and pedagogical quality of module content;
- Effectiveness of the practical experience component;
- Overall satisfaction and learning outcomes.

Each partner country will administer the questionnaire in its national language to maximise response quality and accessibility.

4.4 Reporting of Findings

Upon completion of the evaluation, findings will be documented at two levels:

National Reports (PR8): Each partner will prepare a national report using the template provided in Annex I. These reports will synthesise quantitative survey data and qualitative feedback collected from all participant categories within their country.

Global Summative Report: IPSantarém (SPU), as WP4 lead, will consolidate all national reports into a single summative report, which will present cross-national comparative analysis, overall findings, and evidence-based recommendations for course improvement.

4.5 Course Improvement Process

The findings from the pilot evaluation will be used to iteratively improve the GreenToC course content and platform prior to final dissemination. Each partner is responsible for reviewing recommendations pertaining to the module(s) under their editorial responsibility and implementing scientifically validated revisions. Updated content should be submitted to the shared project repository within the deadline established in the WP4 timeline.

4.6 Instructions for Completing the Evaluation Survey

The following instructions should be communicated to all pilot participants prior to completing the evaluation questionnaire:

Dear Participant,

Thank you for your participation in the pilot testing of the GreenToC course on green transformation strategy development. Your feedback constitutes a critical source of evidence for the scientific and pedagogical improvement of this course. Please follow the instructions below:

Accessing the Survey

- Access the survey via the link provided by your country coordinator: [add national survey link].
- For the optimal experience, use a computer or tablet with an updated web browser (e.g., Chrome, Firefox, Safari).

Preparing to Complete the Survey

- Review the course modules you have attended or facilitated before beginning the survey.
- If you participated in the practical experience component, please reflect on that experience before completing Section D.
- Allow approximately 20–30 minutes to complete the survey thoughtfully.

Answering Questions

- For Likert-scale items (1–5): select the rating that most accurately reflects your experience or assessment.
- For open-ended questions: provide substantive, constructive responses with specific examples where possible.
- Please be candid: your critical observations are as valuable as positive assessments.

After Completing the Survey

- Review your responses before submitting.
- Your responses will be treated with strict confidentiality and used exclusively for research and course improvement purposes.
- You may be contacted by your national coordinator for follow-up clarification.

5. References

Crossman, A. (2019). Pilot study in research. ThoughtCo. Retrieved from <https://www.thoughtco.com/pilot-study-3026449>

European Commission. (2019). The European Green Deal. COM(2019) 640 final. European Commission.

Fraser, J., Fahlman, D., Arscott, J., & Guillot, I. (2018). Pilot testing for feasibility in a study of student retention and attrition in online undergraduate training programs. *International Review of Research in Open and Distributed Learning*, 19(1), 260–278.

Ismail, N., Kinchin, G., & Edwards, J.-A. (2018). Pilot study, does it really matter? Learning lessons from conducting a pilot study for a qualitative PhD thesis. *International Journal of Social Science Research*, 6(1), 1–17.

Mayne, J. (2015). Useful theory of change models. *Canadian Journal of Program Evaluation*, 30(2), 119–142.

Teijlingen, E. R. van, & Hundley, V. (2001). The importance of pilot studies. *Social Research Update*, 35, 1–4.

United Nations. (2015). Transforming our world: The 2030 Agenda for Sustainable Development. A/RES/70/1. United Nations General Assembly.

Annex I – Pilot Testing National Report Template

GreenToC – WP4 Pilot Testing

National Report

[Partner Institution Name]

1. Partner Information

Please provide a brief institutional profile of the partner organisation responsible for this report, including name, country, and relevant expertise in green transformation and/or higher education.

2. Participant Profile Summary

Please present aggregate data on the participants involved in the pilot testing within your country, including: number of students, number of labour market representatives and their organisational sectors, and number of academics involved in course delivery.

3. Summary of Evaluation Results

Please present a summary of the quantitative results from the evaluation questionnaire (mean scores per dimension: platform, module content, practical experience, overall assessment). Include a brief interpretive commentary.

4. Qualitative Feedback: Platform

Please summarise the key suggestions collected from participants for the improvement of the GreenToC learning platform.

5. Qualitative Feedback: Course Content

Please summarise the key suggestions collected from participants for the improvement of the course module content (Modules 1, 2, and 3).

6. Feedback on the Practical Experience Component

Please summarise the feedback from students and labour market representatives regarding the practical experience component, including any challenges encountered and recommendations for improvement.

7. Conclusions and Recommendations

Please present the main conclusions arising from the national pilot and provide evidence-based recommendations for course and platform improvement.

Annex II – WP4 Evaluation Questionnaire

GreenToC – Evaluation of the Green Transformation Course

WP4 Pilot Testing Questionnaire

Dear Participant,

The GreenToC project (Synergies between Higher Education Institutions and the Labour Market for Developing Sustainability Capabilities using Theory of Change and AI) is an Erasmus+ Cooperation Partnership in Higher Education (KA220-HED). The project consortium is inviting you to evaluate the GreenToC semester course on green transformation strategy development, which you have recently attended or facilitated.

Your assessment is of paramount scientific and pedagogical importance: data collected through this instrument will be rigorously analysed and used to improve the course prior to its final dissemination across partner countries. All responses are strictly confidential and will be used exclusively for research and course development purposes.

We thank you in advance for your valuable contribution.

The GreenToC Consortium

SECTION A – Participant Profile

Please indicate your role in the GreenToC pilot course:

- Student (HEI undergraduate/postgraduate)
- Academic / Course instructor
- Labour market representative / Organisation employee

A1. Name of your institution/organisation: _____

A2. Country: _____

Co-funded by



A3. Field of study or professional sector: _____

A4. Prior knowledge of green transformation / sustainability (please tick):

- None
- Basic
- Intermediate
- Advanced

SECTION B – Evaluation of the GreenToC Platform

Please rate the following statements on a scale of 1 (Strongly Disagree) to 5 (Strongly Agree):

B1. The GreenToC platform is easy to navigate and user-friendly.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

B2. The platform is accessible across different devices (computer, tablet, smartphone).

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

B3. Technical issues were minimal during my interaction with the platform.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

B4. Course materials and resources were clearly organised on the platform.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

B5. The platform adequately supports collaborative activities and interaction.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

B6. Please provide any suggestions for improvement of the GreenToC platform:

SECTION C – Evaluation of Module Content

Please select the module(s) you are evaluating (tick all that apply):

- ☐ Module 1 – Green Transformation & Climate Change
- ☐ Module 2 – Theory of Change (ToC) Methodology
- ☐ Module 3 – Digital & AI Technologies for Green Strategies

For each module evaluated, please rate the following statements (1 = Strongly Disagree; 5 = Strongly Agree):

C1. The learning objectives of the module are clearly defined.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C2. The content is scientifically rigorous and up-to-date.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C3. The content is relevant to real-world green transformation challenges.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C4. The instructional materials (readings, videos, exercises) are of high quality.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C5. The module effectively integrates Theory of Change (ToC) concepts.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C6. The module effectively integrates digital and/or AI tools.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C7. The activities and exercises promote active learning.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C8. The workload is appropriate for the allocated time.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

C9. What specific aspects of the module content would you suggest improving? Please provide examples:

SECTION D – Evaluation of the Practical Experience Component (Students & Labour Market Representatives)

(This section is to be completed by students and labour market representatives only)

D1. The practical experience component was well integrated with the theoretical content of the course.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

D2. The organisation I worked with was engaged and supportive throughout the collaboration.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

D3. I was able to apply Theory of Change (ToC) methodology in the development of the green transformation strategy.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

D4. I was able to utilise digital/AI tools in the development of the green transformation strategy.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

D5. The practical experience enhanced my competencies for the labour market.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

D6. The green transformation strategy developed for the organisation is applicable and realistic.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

D7. What were the main challenges encountered during the practical experience?

SECTION E – Overall Assessment of the GreenToC Course

E1. The course met my initial expectations.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

E2. The course provided me with new knowledge and competencies in green transformation.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

E3. The integration of ToC methodology added pedagogical value to the course.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

E4. The use of AI and digital technologies enhanced the learning experience.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

E5. I would recommend this course to colleagues or other students.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

E6. The course contributes meaningfully to the EU Green Deal and sustainability objectives.

1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐

E7. What was the most valuable aspect of this course?

E8. What improvements would you recommend for future iterations of the course?

E9. Additional comments:

Thank you for your contribution to the GreenToC project.